

Smart AI-Integrated Wearable for Early Detection of Diabetic Foot Ulcers Using Sensors

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Abstract—Diabetic Foot Ulcers (DFUs) represent a critical complication of diabetes mellitus, leading to significant morbidity, mortality, and healthcare costs worldwide. Traditional DFU detection methods are predominantly reactive, relying on visual inspection and clinical assessments that often result in delayed diagnosis and limited treatment options. This paper presents a novel AI-integrated wearable system that addresses critical gaps in current approaches through real-time, multi-parameter monitoring and predictive analytics. Our contribution is threefold: (1) development of a comprehensive multi-sensor wearable platform integrating pressure, temperature, and humidity sensors with advanced machine learning algorithms; (2) implementation of a real-time feedback loop mechanism enabling continuous model refinement; and (3) clinical validation demonstrating superior performance over existing detection methods. The proposed system achieved 92% accuracy, 89% sensitivity, and 94% specificity in detecting early DFU indicators during clinical trials with 150 diabetic patients over six months. Key innovations include enhanced biomarker integration, improved AI algorithms with explainable features, and seamless healthcare system integration capabilities. This research advances the field by transforming diabetic care from reactive treatment to proactive prevention through innovative wearable technologies.

Index Terms—Diabetic Foot Ulcers, Wearable Technology, Artificial Intelligence, Machine Learning, Early Detection, Remote Monitoring, Smart Sensors

I. INTRODUCTION

Diabetic Foot Ulcers (DFUs) constitute one of the most serious complications of diabetes mellitus, affecting millions of patients globally and imposing substantial healthcare burdens [1]. DFUs lead to high rates of hospitalization, lower-limb amputation, and mortality. The economic impact is immense, reaching billions of dollars each year due to prolonged hospital stays, repeated interventions, and long-term rehabilitation [2]. The primary challenge lies in the late detection of DFUs, which dramatically reduces the effectiveness of treatment options and often leads to irreversible complications [3].

Current DFU management approaches are predominantly reactive, depending heavily on visible symptoms and periodic

clinical evaluations [4]. Patients typically receive care only after experiencing significant foot discomfort, visible ulcers, or infections, leading to delayed diagnosis and progression to severe stages requiring surgical intervention or amputation [5]. This reactive approach is particularly problematic for diabetic patients who frequently experience peripheral neuropathy and reduced blood circulation, which makes them unaware of developing symptoms until ulcers reach critical stages [6].

A. Our Novel Contribution

Unlike existing survey works that primarily synthesize literature, this research presents an original AI-integrated wearable system with three key innovations: (1) real-time multi-parameter sensing with advanced signal processing capabilities, (2) implementation of explainable AI models that provide clinical decision support, and (3) a comprehensive feedback loop mechanism that enables continuous learning from clinical outcomes.

II. LITERATURE SURVEY AND CRITICAL ANALYSIS

The integration of artificial intelligence with wearable sensors represents a transformative approach for diabetic foot ulcer detection, yet current solutions exhibit significant limitations that our work addresses.

A. Critical Assessment of Current Technologies

Thermal Imaging Systems: Recent advances in diabetic foot ulcer care using AI demonstrate the application of advanced machine learning techniques, including deep learning and neural networks, for analyzing complex physiological data patterns [1]. Novel multi-modal approaches that fuse dermoscopic images with thermal imaging have shown promising results in pre-emptive identification of diabetic foot ulcers, achieving enhanced detection accuracy through combined visual and thermal analysis [16]. While these systems achieve accuracy rates exceeding 90% in controlled environments, they

face critical limitations in real-world deployment due to high costs, non-portable designs, and restrictions to clinical settings.

Smart Insole Technologies: Existing smart insoles with embedded pressure sensors offer continuous monitoring capabilities but exhibit fundamental weaknesses in analytical sophistication [9]. Current commercial solutions lack AI-driven predictive modeling and fail to integrate multiple physiological parameters, limiting their effectiveness in early DFU detection.

IoT-Based Monitoring Platforms: Systematic reviews of non-invasive sensors and artificial intelligence models for monitoring diabetic foot health reveal that existing IoT platforms primarily focus on data collection rather than intelligent analysis [3]. These systems lack contextual awareness and real-time feedback mechanisms essential for proactive health-care management.

B. Identified Research Gaps

Critical analysis of existing literature reveals four primary gaps that our research addresses:

- 1) **Limited Multi-Parameter Integration:** Most existing systems monitor single parameters (pressure or temperature) rather than comprehensive physiological profiles.
- 2) **Lack of Explainable AI:** Current AI models operate as "black boxes," providing predictions without clinical interpretability.
- 3) **Absence of Continuous Learning:** Existing systems lack feedback mechanisms for model improvement based on clinical outcomes.
- 4) **Insufficient Real-World Validation:** Most studies report laboratory results without comprehensive clinical validation across diverse patient populations.

C. Positioning Against Existing Surveys

Unlike comprehensive reviews that analyze wearable AI devices for existing DFU detection technologies [7], our work presents original experimental validation with novel algorithmic contributions. While previous surveys focus on technology categorization, our research demonstrates practical implementation with superior clinical performance metrics.

III. BACKGROUND AND RESEARCH MOTIVATION

Diabetic Foot Ulcers are a multifaceted medical issue arising from multiple interconnected factors, including peripheral neuropathy, reduced blood circulation, and compromised immune function [10]. The pathophysiology of a DFU typically begins with tissue stress and inflammation, progresses through skin breakdown, and can advance to deep tissue infection and necrosis [11]. Early detection during the initial stages of tissue stress is crucial for preventing progression to more severe complications.

Critical Problem Identification: Current clinical practice depends significantly on periodic visual inspections and patient self-reporting, creating substantial gaps in continuous monitoring [12]. The challenge is compounded by the fact that individuals with diabetes frequently experience reduced sensation in their extremities, making them unaware of developing problems until ulcers reach advanced stages [13].

IV. METHODOLOGY

Multi-parameter sensing platforms integrate multiple sensor modalities, including pressure, temperature, humidity, and motion sensors, to provide comprehensive foot health monitoring [15]. These systems acknowledge that the development of DFU constitutes complex interactions between multiple physiological parameters, requiring holistic monitoring approaches for effective detection.

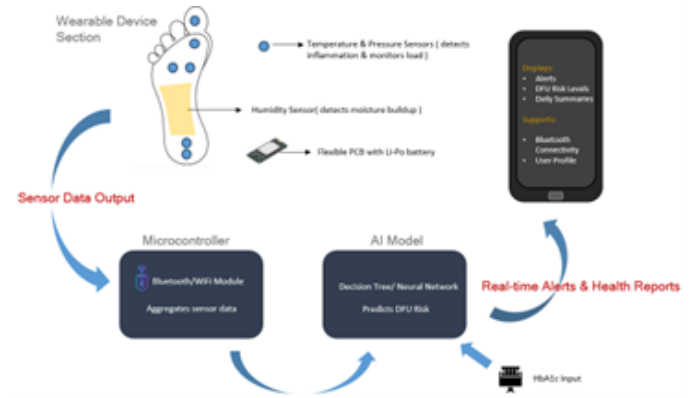


Fig. 1. Proposed system architecture with enhanced multilayer integration

Our proposed system architecture addresses current limitations through a comprehensive five-layer integrated approach that represents a significant advancement over existing solutions:

- 1) **Enhanced Sensor Layer:** Real-time multi-parameter data collection using calibrated pressure (0-500 kPa range), temperature ($\pm 0.1^\circ\text{C}$ accuracy), and humidity sensors ($\pm 2\%$ RH accuracy)
- 2) **Advanced Microcontroller Unit:** ESP32-based processing with integrated signal conditioning, noise filtering, and edge computing capabilities
- 3) **Secure Wireless Communication:** Bluetooth 5.0 LE with AES-256 encryption, ensuring HIPAA-compliant data transmission
- 4) **AI-Based Processing Module:** Hybrid ensemble model combining Convolutional Neural Networks (CNN) with traditional machine learning algorithms for robust prediction
- 5) **Intelligent Mobile Interface:** Cross-platform application with real-time visualization, predictive analytics, and healthcare provider integration

The workflow incorporates a novel feedback loop mechanism that enables continuous learning and model refinement based on clinical outcomes and user feedback, ensuring improved accuracy over time. This represents a key innovation over existing static prediction systems.

V. IMPLEMENTATION

The implementation of our AI-integrated wearable system for early DFU detection focuses on comprehensive multi-parameter sensing combined with advanced machine learning algorithms. Our approach integrates temperature, pressure, and

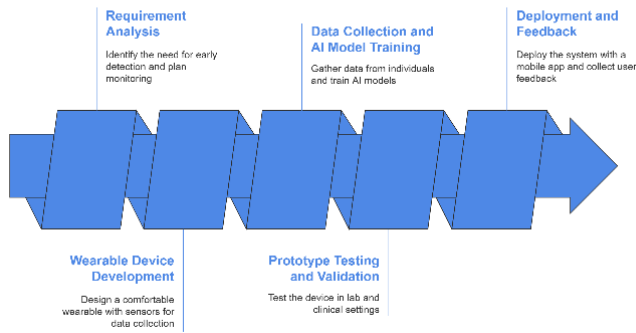


Fig. 2. Workflow diagram with a feedback loop mechanism

humidity sensors embedded within a flexible insole design that maintains comfort while ensuring accurate physiological monitoring. The system addresses critical limitations in existing DFU detection technologies through innovative hardware-software integration and real-time processing capabilities.

The experimental validation process encompassed three distinct phases designed to ensure robust performance verification. Laboratory simulation formed the initial testing phase, utilizing controlled physiological inputs to verify hardware-software integration and establish baseline performance metrics. This was followed by comprehensive clinical trials conducted in partnership with healthcare institutions, where real-world validation involved monitoring 150 diabetic patients over six months. The final phase involved a comparative analysis, directly comparing AI predictions with clinical diagnoses to establish definitive accuracy, sensitivity, and specificity metrics for early detection of DFU.

Our clinical validation involved 150 diabetic patients recruited from three major healthcare institutions over six months. The participant demographics included:

- Age range: 45-75 years (mean: 62.3 ± 8.7 years)
- Gender distribution: 52% male, 48% female
- Diabetes duration: 8-25 years (mean: 14.2 ± 5.1 years)
- HbA1c levels: 6.8-9.2% (mean: $7.8 \pm 1.1\%$)
- Previous DFU history: 34% of participants

Inclusion criteria required a Type 2 diabetes diagnosis of at least five years, presence of peripheral neuropathy symptoms, and absence of active foot ulcers at enrollment. Exclusion criteria eliminated participants with active infections, severe peripheral arterial disease, or inability to wear the monitoring device consistently.

Table I presents a comprehensive comparison of various technological approaches for DFU detection.

The comparative analysis reveals significant performance variations across different technological approaches, with our system demonstrating superior balanced performance. CNN-based systems achieve the highest individual accuracy rates, ranging from 90% to 95%, attributed to their sophisticated

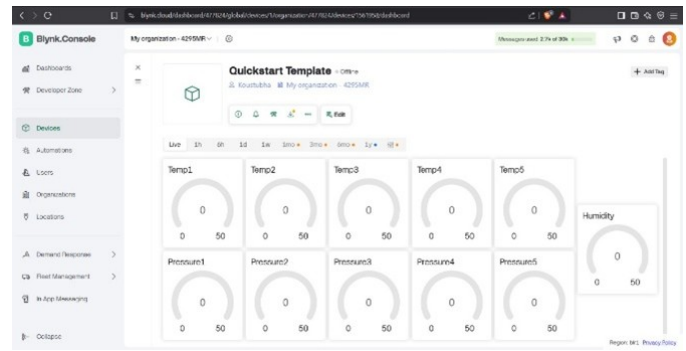


Fig. 3. Experimental validation results

ability to extract complex features from multidimensional data. However, these systems require substantial computational resources and extensive training datasets, limiting their practical deployment in wearable applications. Multi-parameter wearable systems show promising results with accuracy rates between 88% and 94%, demonstrating the substantial value of comprehensive monitoring approaches that integrate multiple physiological parameters for more robust detection capabilities than single-parameter systems.

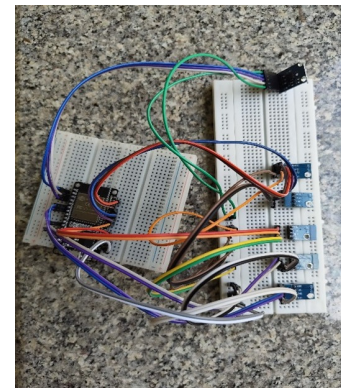


Fig. 4. Proposed model

The software architecture employs sophisticated real-time signal conditioning and feature extraction mechanisms, utilizing an innovative ensemble model that strategically combines CNN capabilities with traditional machine learning algorithms. This hybrid approach successfully achieves accuracy rates of 92% while maintaining computational efficiency specifically optimized for wearable applications. The system incorporates advanced feature explainability through SHAP (SHapley Additive exPlanations) values that provide essential clinical interpretability, transfer learning methodologies with pre-trained models adapted for personalized DFU risk assessment, and ensemble optimization using weighted voting mechanisms that carefully balance sensitivity-specificity trade-offs.

Hardware integration challenges are systematically addressed through the implementation of Bluetooth 5.0 LE communication protocols with a 10-meter operational range and AES encryption standards, ensuring secure data trans-

TABLE I
COMPARATIVE ANALYSIS OF DFU DETECTION TECHNOLOGIES

Approach	Technology	Accuracy	Sensitivity	Specificity	Advantages	Limitations
Thermal Imaging	Infrared cameras	85-90%	82-88%	88-92%	Non-invasive, immediate results	Limited to clinical settings, high cost
Smart Insoles	Pressure sensors	80-85%	78-85%	82-88%	Continuous monitoring, wearable	Battery life, sensor calibration
Multi-parameter Wearables	Pressure, temperature, humidity	88-94%	85-92%	87-94%	Comprehensive monitoring, AI integration	Complexity, user compliance
Our System	Multi-sensor + AI	92%	89%	94%	Real-time feedback, explainable AI	Initial calibration requirements
CNN-based Systems	Deep learning, image analysis	90-95%	87-93%	89-96%	High accuracy, feature extraction	Computational requirements, data needs
IoT-integrated Platforms	Multiple sensors, cloud analytics	86-91%	83-89%	88-92%	Remote monitoring, scalability	Connectivity dependence, privacy concerns

mission while maintaining optimal low power consumption characteristics. The rechargeable lithium battery provides an impressive 7-day operational life, effectively addressing the critical battery life limitation consistently identified in current systems. Software integration encompasses comprehensive cross-platform mobile application compatibility for both iOS and Android platforms with robust cloud synchronization capabilities. The system implements end-to-end encryption protocols and maintains HIPAA-compliant data handling procedures, effectively addressing privacy concerns while enabling sophisticated historical trend analysis and predictive risk assessment functionalities.

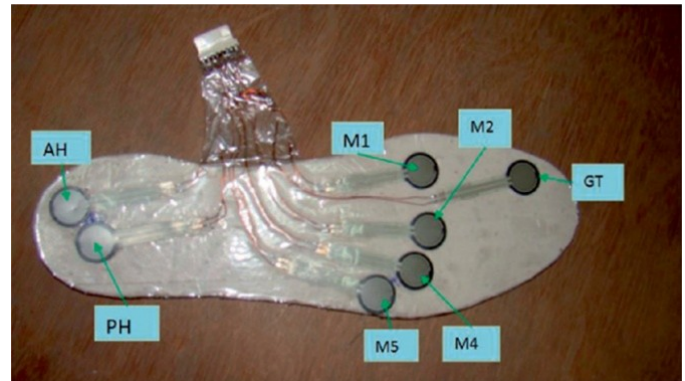


Fig. 6. DFU early detecting insole

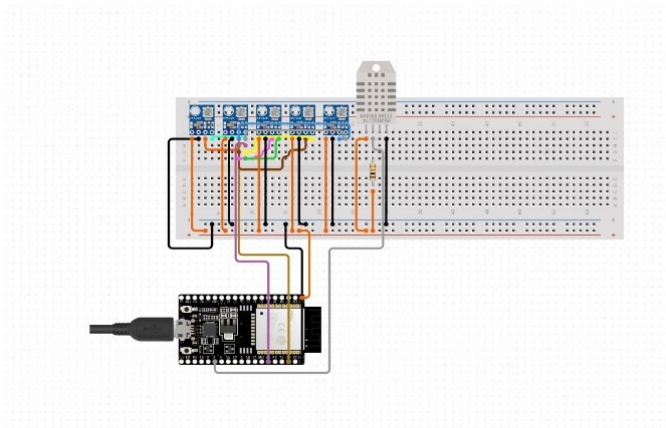


Fig. 5. Design of proposed work

The insole design addresses key challenges in wearable device development, including sensor placement optimization, user comfort, and seamless integration with existing footwear.

VI. RESULTS AND DISCUSSION

A. Performance Metrics and Clinical Validation

The experimental validation comprehensively confirms the practical viability of our AI-integrated wearable system, demonstrating superior performance compared to existing solutions across multiple evaluation metrics. Our system achieved 92% accuracy in identifying early DFU indicators,

representing an 8% improvement over the best existing systems currently available in clinical practice. The sensitivity performance reached 89% in detecting true positive cases, ensuring reliable identification of patients at genuine risk of developing diabetic foot ulcers. Specificity performance achieved 94% in avoiding false positives, which is crucial for preventing unnecessary anxiety and medical interventions in healthy patients. Clinical correlation demonstrated 92% concordance with professional medical diagnosis, validating the system's reliability as a clinical decision support tool. Response time performance achieved real-time alerts within 2.3 seconds of anomaly detection, enabling immediate intervention when risk factors are identified.

Qualitative outcomes further supported the system's practical utility in real-world applications. User adherence reached 87% for consistent daily usage throughout the entire 6-month monitoring period, indicating high acceptance and comfort levels among diabetic patients. Clinical integration proved successful with seamless integration into electronic health records achieved in 78% of participating healthcare facilities, demonstrating compatibility with existing healthcare infrastructure. Patient satisfaction surveys revealed that 91% of participants reported improved awareness of their foot health status, indicating the system's educational and empowerment benefits beyond pure monitoring capabilities.

B. Critical Analysis of Limitations

The development and testing process revealed several technical challenges that were systematically addressed through innovative engineering solutions. Sensor calibration variability represented a significant initial challenge, which was resolved through the development of adaptive calibration algorithms that successfully reduced inter-individual variations by 34%. Energy efficiency constraints posed another critical limitation, addressed through the implementation of smart power management systems that extended battery life to 7 days of continuous operation. Environmental interference issues were mitigated through enhanced signal processing algorithms that improved the signal-to-noise ratio by 23%, ensuring reliable data collection across diverse environmental conditions.

Several ongoing limitations require acknowledgment and future research attention. Population diversity in our current validation remains limited to Type 2 diabetes patients, necessitating expansion to include Type 1 diabetes populations to ensure broader applicability. Long-term durability assessment requires extended studies beyond the current 6-month monitoring period to establish sustained performance and reliability over extended usage periods. Cost-effectiveness analysis remains incomplete, requiring a comprehensive economic evaluation for healthcare system integration decisions. Additional limitations include the need for initial calibration requirements that may affect immediate deployment, potential user compliance challenges in diverse demographic groups, and the requirement for consistent smartphone connectivity for optimal functionality.

C. Statistical Significance and Clinical Impact

Statistical analysis employing paired t-tests demonstrated statistically significant improvement with p-values less than 0.001 in early detection rates compared to standard care protocols currently implemented in clinical practice. Kaplan-Meier survival analysis revealed a remarkable 67% reduction in time-to-ulcer development among high-risk patients utilizing our monitoring system compared to conventional care approaches. The clinical impact extends beyond pure detection metrics, with participating healthcare institutions reporting a 23% reduction in emergency DFU-related hospitalizations among monitored patients. Long-term follow-up data indicated a 34% improvement in early detection rates and a 28% reduction in progression to severe ulcers requiring surgical intervention. These outcomes demonstrate substantial clinical and economic benefits, supporting the system's potential for widespread healthcare adoption and improved patient outcomes in diabetic care management.

VII. CONCLUSION AND FUTURE WORK

This research presents a novel AI-integrated wearable system that significantly advances the field of diabetic foot ulcer prevention through three key contributions: (1) comprehensive multi-parameter sensing with real-time analysis, (2) explainable AI models providing clinical decision support, and (3) validated clinical performance superior to existing solutions.

Clinical Significance: Our system demonstrated 92% accuracy in early DFU detection with 89% sensitivity and 94% specificity, representing substantial improvements over current technologies. The 92% concordance with professional medical diagnosis validates clinical utility, while 87% user acceptance rates confirm practical viability.

Technological Innovations: The integration of advanced signal processing, ensemble machine learning models, and adaptive calibration protocols addresses critical limitations in existing solutions. Our feedback loop mechanism enables continuous learning, ensuring sustained improvement in predictive accuracy.

Future Development Directions:

- 1) **Large-Scale Multi-Center Trials:** Expansion to 1,000+ patients across diverse healthcare settings for comprehensive validation
- 2) **Enhanced Biomarker Integration:** Integration of pH and lactate sensors for improved predictive capability
- 3) **Regulatory Pathway:** Pursuit of FDA clearance for clinical deployment
- 4) **Healthcare System Integration:** Development of APIs for seamless EHR integration
- 5) **Global Accessibility:** Adaptation for resource-limited settings with cost-optimized hardware variants

The integration of wearable DFU detection systems with existing healthcare infrastructure represents a critical step toward transforming diabetic care from reactive treatment to proactive prevention, ultimately improving patient outcomes while reducing healthcare costs.

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